Nanotechnology and Quantum Applications special laboratory 2

Code: BMETEFTBsPSPL2-00
Language: English
Location: BME,
Responsible: Szabolcs Csonka, András Halbritter
Lab leaders / supervisors:
I ODMR (ODMR) / Ferenc Simon (simon.ferenc@ttk.bme.hu), Bence Göblyös(bgoblyos@edu.bme.hu)
II High Tc SQUID (SQUID) / Olivér Kürtössy (o.kurtossy@gmail.com) , Peter Makk (makk.peter@ttk.bme.hu)
III Entangled photon pairs (EntPhoton) / Csaba Holló (hollo.csaba@ttk.bme.hu), Tamás Sarkadi (sarkadi.tamas@ttk.bme.hu)
IV Ultrafast Laser pulses (FastPulse) / Pál Maak (maak.pal@ttk.bme.hu)

Requirements

Attendance requirements: It is compulsory to execute
Option a) 4 measurements or
Option b) 3 measurements plus presentation of own research topic. Measurements are 2x4hours long. In case of **certified absence**, a maximum of 1 replacement measurement (altogether 2x4 hours) are possible at a time agreed by the tutors.
Option c) like in Option b), except that presence at all 4 measurements, but lab report is required only from 3 measurements.

Working method: the measurements are executed in pairs. Please indicate your preferred pairs for the measurements in this Excel file by 22nd February. Option b) is only possible if both members of the pair choose it and a written certificate is required from the research supervisor that the student is actively carrying out research work.

Schedule and place of the measurements (the numbers in the table denote the number of a measurement pair, see the pair codes here):

Time-period of the courses:
Team 1: Wednesday 8:30-12:00
Team 2: Wednesday 13:15-17:00

A lab responsible will be present continuously during these times. Breaks are not scheduled, but you may take a break and leave the lab as needed during measurements.

Measurements exercises (see in lab descriptions):

- I ODMR - (place: F32L2 Physics Student Lab, FIII building 2nd floor)
- II SQUID - (place: FI building, Basement, Nanoelectronics lab)
- III EntPhoton - (place: FIII Building, Basement, Dep. Of Atomic Physics)
- IV FastPulse - (place: F III Building, Dep. Of Atomic Physics)

Team 1 (8pairs)

Week	**2**	**3**	**4**	**5**	**6**	**7**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
Date	2/25	3/4	3/11	3/18	3/25	04/01	04/22	04/29	05/06	05/13	05/20	05/27	
ODMR	1,2,3	1,2,3	7,8	7,8	4,5,6	4,5,6							
SQUID	4,5,6	4,5,6			1,2,3	1,2,3	7,8	7,8					
EntPhoton	7,8	7,8	1,2	1,2			3,6	3,6	4,5	4,5			
FastPulse			4,5	4,5			1,2	1,2	7,8	7,8	3,6	3,6	
Presentation									1,2	3,6	4,5	7,8	

Team 2 (10pairs)

Week	2	3	4	5	6	7	9	10	11	12	13	14	15
Date	2/25	3/4	3/11	3/18	3/25	04/01	04/22	04/29	05/06	05/13	05/20	05/28	
ODMR	1,2,3	1,2,3	7,8,9	7,8,9	4,5,6	4,5,6	10	10					
SQUID	4,5,6	4,5,6	10	10	1,2,3	1,2,3	7,8,9	7,8,9					
EntPhoton	9,10	9,10	1,2	1,2	7,8	7,8	3,6	3,6	4,5	4,5			
FastPulse			4,5	4,5	9,10	9,10	1,2	1,2	7,8	7,8	3,6	3,6	
Presentation									10, 1	3,6,2	4,5,9	7,8	

Grading:

Measurements

- Entrance test (30 points) at the start of the first session.
- Lab notes (30 points total), submitted at the end of each session.
- Lab report (40 points), due within 1 week of the second session.

Presentation

- Option a) 1 oral presentation at the end of the semester on one of the measurements (100points) in 20 minutes with 10minutes discussion). Evaluation form is used (Scores for quality of a) structuring the content, b) introduction, c) presentation of the methodology, d) presentation of own results, e) adequate conclusion, f) visual quality, clarity of speech, confidence, formal aspects, ...).
- Option b) and c) Like in Option a) plus:
  - One more oral presentation on the own research work (50points)
  - In case the student has participated at TDK conference, no need for oral presentation (50point is automatic). 1 TDK presentation only allows skipping 1 oral presentation either in Spec. lab 1 or 2).

Total Points:

- Option a): 500 points
- Option b, c): 450 points

Grade Scale:

- <40%: Fail (1)
- 40%-55%: Pass (2)
- 55%-70%: Satisfactory (3)
- 70%-85%: Good (4)
- 85%-100%: Excellent (5)

Note: The failure criterion applies to the overall score, not to individual components (entrance tests, lab notes, or reports). All measurements must be completed, but a low score in one area will only lower the aggregate grade without requiring a replacement measurement.

Submission Rules:

- **Entrance Tests:** Must be completed at the begining of the class. Late arrivals will have less or no time, leading to a potential loss of points.
- **Lab Notes:** Submit online in Moodle within 1 hour of the end of each session. Late submission results 10-point-deduction per hour.
- **Lab Reports:** Must be submitted within one week (7x24 hours) from the start of the second lab session. A one-time, one-week extension is allowed without penalty. Any further delay results in a 10-point-deduction per week, until the score for the lab report reaches zero.

Lab leaders should correct the lab reports within one week from the submission deadline, with a single possibility for an extra week delay. Please complain, if this is not satisfied.

Student Pairs:

Morning	Name1	Neptun code	Name 2	Neptun code	Option A, B, C	Skipped measurement
1	Mukli Márton Gábor	FDEQBX	Tompos Gábor	PYT84Z	B	Fast Pulse
7	Ladányi Márton		Kohut Márk Balázs	IR173M		

2	Árok Anna	BJF7AG	Éberhardt Milán	F5U7ZL	A	
3	Beke Bálint	OYRSP2	Karácsonyi Levente	Y3BW2Z	A	
4	Karminás Bence	SAGXYK	Bucskó Gergely	GWCLHO	A	
5	Varga Aurél	SVG8LE	Bányai Áron	FAWKVK	A	
6	Szathury Csaba	NBP6LW	Borcsok Balazs	JR0D2G		
7						
8						
9						
10						
11						
Afternoon						
1	Bálint Béla Mária	H31MOD	Csabai Szabolcs Imre	JINF8A	B	
2	Mikes Tamás Kelemen	BJPB5C	Szebedy András Pál	HS2M9E	A	
3	Harmos Imre	DO7RDH	Hunyady Csongor		B	Fast Pulse
3	Bence Göblyös	CM5PHB	Will join as 3rd to other groups as possible	N/A	B	ODMR
4	Jaca Gregorio	U8L9B9	Tállosy Péter	K14WR1	A	
5	Ajzenberger Lea	QK2Y4Z	Volk János	KKY30Q	B	Fast Pulse
6	Csoma Zsófia	KFHXBR	Márton Kornél	OI7V8R	A	
7	Bajnóczi Boldizsár	S9BYHX	Török Barnabás	FY16Y2	A	
7	Tichy Márk		Join as 3rd partner.			
8	Wagner Andor	JBCDBJ	Szontagh Bence	JJHFL3	B	Fast Pulse
9	Yuri Fujii	OPX6W0	Zakariás Kozma		A	

Oral presentations:

In case you chose Option B or C:
- you have to prepare 2nd talk about your own research - or prove that you presented it at the TDK conference.
- in case you had no TDK presentation please bring a written certificate from your supervisor that you are working with him/her .

Places
Semroom - Department of Physics, F building I, 1st floor, Room 10

Program

Week 11 06 May

Student pair	Name	Topic	Time	Place	In case own research topic was presented already at TDK conference 2025, give its title.

1	1	ODMR	9:00	Semroom	
1	2	SQUID	9:30	Semroom	
1	1	Own Research	10:00	Semroom	
1	2	Own Research	10:30	Semroom	
2	1	EntP	11:00	Semroom	
2	2	ODMR	11:30	Semroom	

Student pair	**Name**	**Topic**	**Time**	**Place**	In case own research topic was presented already at TDK conference 2025, give its title.
10	1	ODMR	14:15	Semroom	
10	2	EntP	14:45	Semroom	
1	1	FastP	15:15	Semroom	
1	2	SQUID	15:45	Semroom	
1	1	Own Research	16:15	Semroom	
1	2	Own Research	16:45	Semroom	

Week 12 13th May

Student pair	**Name**	**Topic**	**Time**	**Place**	In case own research topic was presented already at TDK conference 2025, give its title.
3	1	ODMR	9:00	Semroom	
3	2	SQUID	9:30	Semroom	
6	1	EntP	10:00	Semroom	
6	2	FastP	10:30	Semroom	
			11:00	Semroom	
			11:30	Semroom	

Student pair	**Name**	**Topic**	**Time**	**Place**	In case own research topic was presented already at TDK conference 2025, give its title.
3	1	ODMR	14:15	Semroom	
3	2	SQUID	14:45	Semroom	
6	1	EntP	15:15	Semroom	
6	2	FastP	15:45	Semroom	
2	1	ODMR	16:15	Semroom	
2	2	EntP	16:45	Semroom	
3	3	SQUID	17:15	Semroom	

Week 13 20th May

Student pair	**Name**	**Topic**	**Time**	**Place**	In case own research topic was presented already at TDK conference 2025, give its title.
4	1	ODMR	9:00	Semroom	
4	2	SQUID	9:30	Semroom	
7	1	SQUID	11:00	Semroom	
7	2	ODMR	11:30	Semroom	

Student pair	**Name**	**Topic**	**Time**	**Place**	In case own research topic was presented already at TDK conference 2025, give its title.
4	1	ODMR	14:15	Semroom	
4	2	SQUID	14:45	Semroom	
5	1	EntP	15:15	Semroom	
5	2	FastP	15:45	Semroom	
9	1	ODMR	16:15	Semroom	
9	2	SQUID	16:45	Semroom	
5	1	Own research	17:15	Semroom	
5	2	Own research	17:45	Semroom	

Week 14 27th May
Afternoom

Student pair	**Name**	**Topic**	**Time**	**Place**	In case own research topic was presented already at TDK conference 2025, give its title.
5	1	ODMR	9:00	Semroom	
5	2	SQUID	9:30	Semroom	

Student pair	**Name**	**Topic**	**Time**	**Place**	In case own research topic was presented already at TDK conference 2025, give its title.
7	1	EntP	14:15	Semroom	

7	2	FastP	14:45	Semroom	
8	1	ODMR	15:15	Semroom	
8	2	SQUID	15:45	Semroom	
8	1	Own Research	16:15	Semroom	
8	2	Own Research	16:45	Semroom	
7	3	SQUID	17:15	Semroom	

References to prepare your talk:

ODMR

[1] The nitrogen-vacancy colour centre in diamond
Doherty et al., *Physics Reports* **528**, 1–45 (2013)
DOI: https://doi.org/10.1016/j.physrep.2013.02.001
OA: https://arxiv.org/abs/1302.3288

[2] Quantum error correction in a solid-state hybrid spin register
Waldherr et al., *Nature* **506**, 204–207 (2014)
DOI: https://doi.org/10.1038/nature12919
OA: https://arxiv.org/abs/1309.6424

[3]· High-sensitivity diamond magnetometer with nanoscale resolution
Taylor et al., *Nature Physics* **4**, 810–816 (2008)
DOI: https://doi.org/10.1038/nphys1075
OA: https://arxiv.org/abs/0805.4783

[4]· Coherence of single spins coupled to a nuclear spin bath of varying density
Balasubramanian et al., *Nature Materials* **8**, 383–387 (2009)
DOI: https://doi.org/10.1038/nmat2420
OA: https://arxiv.org/abs/0811.4464

SQUID

[1] J. S´olyom, Fundamentals of the Physics of Solids (Vol 2)
[2] P. Makk, Transport in complex nanostructures lecture notes
[3] C. Granata et al., Physics Reports, 614,1 (2016)

[4] The Feynman lectures on Physics: The Schr ̈odinger Equation in a Classical Context: A Seminar on Superconductivity
[5] Yong-Joo Doh et al., Science 309, 272 (2005)
[6] Mr. SQUID User's Guide
Further interesting readings
[7]M. Hamalainen et al., Rev. Mod. Phys. 65, 413 (1993)
[8] D. Vasyukov et al., Nature Nanotechnology 8, 639–644 (2013)
[9] D. Halbertal et al., Nature 539, 407–410 (2016)
[10] D. S. Kalashnikov et al. Phys. Rev. B 112, 144504 (2025)

EntPhoton

[becker2005] book on Time-Correlated Single Photon Counting Techniques
[couteau2018] review on SPDC
[hadfield2009] review on single photon detectors
[kwiat1995] article on one of the first entangled photon pair source
Link

FastPulse

[1] Varjú, K., Kovács, A., Kurdi, G. *et al.* High-precision measurement of angular dispersion in a CPA laser. *Appl Phys B* **74** (Suppl 1), s259–s263 (2002). https://doi.org/10.1007/s00340-002-0882-z https://link.springer.com/article/10.1007/s00340-002-0882-z#preview

[2] Á Börzsönyi, PhD theses, 2012, Szegedi Tudományegyetem, https://doktori.bibl.u-szeged.hu/id/eprint/1423/1/BorzsonyiA_ertekezes.pdf https://www.ape-berlin.de/en/autocorrelator/mini-tpa/

[3] W. Demtröder, Laser Spectroscopy Vol.2. , 2015, Springer https://faculty.kfupm.edu.sa/phys/magondal/Phys608/Demtroeder%205th%20edition.pdf

Guildlines to prepare your talk

1) About the structure of the presentation:

On the **title** slide please include your name, the date, and your affiliation / place of research.

Start either by describing the **motivation** to the research, or a contents slide. Show a list of **contents** at some point in the presentation, either before or after the motivation, but definitely before you get deeper into the topic or your results.

The motivation can be as short as a couple of minutes, but don't be too short. It can be as long as 4-6 minutes (depends on the topic and your time limit), but stick to a few things only. It is counter-productive to present lots of examples and lose focus. If you mention something, make sure you know the background at least superficially (look in the references if necessary).
Please **read the References** related to the lab course given in this file (see previous section) and build your motivation and introductory part using the information learnt from them.
Discuss your experimental **methods** and the theoretical **background** at least in passing. If it is complicated but the details are not important, you can skip parts of it and focus on what is relevant later. If they are important, spend at least a few minutes on the basics and also define the physical quantities, so that the audience has some concept of it all.
Structure the talk in your head, and present it accordingly. For example, pause between major parts for a few seconds and then say out loud what comes next. You could even put in an extra slide before each main part of the talk that shows only the title of this next part, in line with the Contents.
At the end, make a **summary** or conclusions slide to go over your main results again. You can even show earlier pictures again. After the conclusions you might mention again some of the practical applications, but do not discuss them in detail. Focus on your achievements, stay positive, and maybe give an outlook and talk about how you plan to continue your research.
Finally, **acknowledge** all important contributors, from fellow researchers who took part to other research groups who provided materials, or institutions where you performed part of your research, or funding projects, grants and fellowships.

2) General advice:
Practice the talk at least a couple of times out loud, and measure the time. Cut off parts as necessary to keep it short enough.
Use **figures** and illustrations, try to avoid slides with text only, especially in the motivation / introduction. If possible, do not describe something and make the audience rely on their imagination, if you can instead illustrate that something by a schematic or a photo. Don't leave half a slide empty (unless it is a title slide); add an illustration or enlarge an existing one (within reason), or merge two slides into one if it is reasonable. If you have data in text or in tables that can also be shown in a figure, make the figure. If a schematic seems old/ugly, you can find a newer one or make one yourself.
Note on the slides the sources and references that you get your material from, especially for images and figures. Add slide numbers.
Please don't use screenshots of computer programs (windows and other user interfaces), at least crop to show only the graph.
Define quantities on the slide: actually write down their names.

Don't write too much text if possible, like long sentences. No need to write down every numerical result. No need to present every piece of research. Focus on a main story and do not get too sidetracked.
You may use notes but don't rely on them too much, otherwise you may get stuck easily. Make the slides so that they are helpful for you, then remember to look up at your slide. And we can turn this around: describe your figures during your talk, if it's up there, you should explain it or just point to it at least once. Use the pointer/mouse.
In evaluating the data, always try to consider what may be the error of the quantity, at least from how much variation there is between repeated measurements. Then consider how many decimal digits in the quantity are useful. For example, if the quantity you read out is 2.0578 and the error (or scatter) is 0.1579, the useful digits in both are roughly two, and then just write 2.06 +- 0.16. Don't estimate relative differences between quantities that are themselves relative: like temperatures in Celsius (it is not an absolute temperature scale, but measured relative to 273 K).
You can mention challenges or failures, but focus on how you overcame them. You can mention if there was something that did not work or that you could not finish, but do not dwell on it and do not collect them in one place. Focus on the results that you have achieved, and try to **stay positive**.